

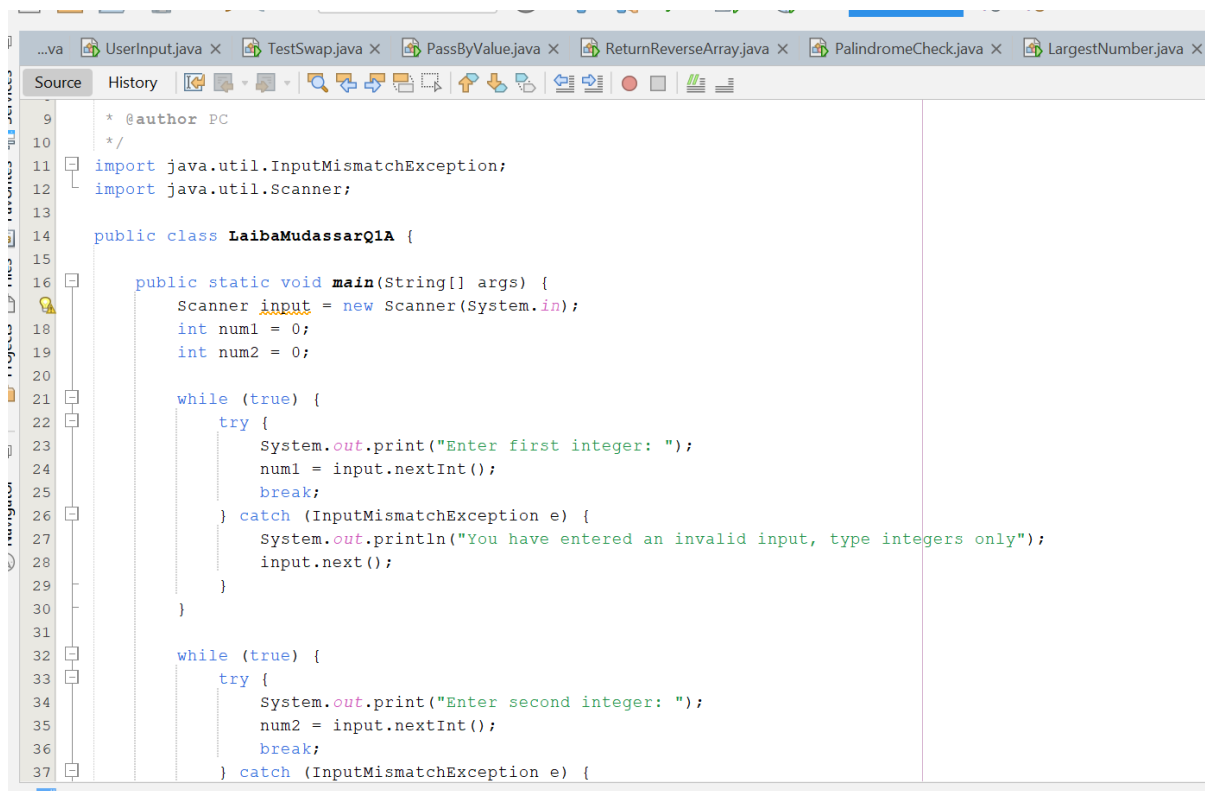
**Laiba Mudassar**

**Sp26-bcs-021**

**Lab assignment 2**

**programming fundamentals**

## Question no 1 (a)



```
9      * @author PC
10     */
11     import java.util.InputMismatchException;
12     import java.util.Scanner;
13
14     public class LaibaMudassarQ1A {
15
16         public static void main(String[] args) {
17             Scanner input = new Scanner(System.in);
18             int num1 = 0;
19             int num2 = 0;
20
21             while (true) {
22                 try {
23                     System.out.print("Enter first integer: ");
24                     num1 = input.nextInt();
25                     break;
26                 } catch (InputMismatchException e) {
27                     System.out.println("You have entered an invalid input, type integers only");
28                     input.next();
29                 }
30             }
31
32             while (true) {
33                 try {
34                     System.out.print("Enter second integer: ");
35                     num2 = input.nextInt();
36                     break;
37                 } catch (InputMismatchException e) {
```

```
while (true) {  
    try {  
        System.out.print("Enter second integer: ");  
        num2 = input.nextInt();  
        break;  
    } catch (InputMismatchException e) {  
        System.out.println("You have entered an invalid input,  
        input.next();  
    }  
}  
  
int sum = num1 + num2;  
System.out.println("The sum is: " + sum);  
input.close();  
}  
}
```

Output - Arrays (run) ×

run:  
Enter first integer: 5  
Enter second integer: 7  
The sum is: 12  
BUILD SUCCESSFUL (total time: 7 seconds)

## Question no 1 (b)

```
Source History
D:\1st_Semester\Programming_Fundamental\practice\Arrays\src\arrays\UserInput.java
package arrays;

6
7
8
9
10
11 /**
12  *
13  * @author PC
14  */
15 import java.util.InputMismatchException;
16 import java.util.Scanner;
17
18 public class LaibaMudassarQ1B {
19
20     public static void main(String[] args) {
21         Scanner input = new Scanner(System.in);
22         int[] numbers = new int[10];
23
24         System.out.println("Enter 10 integer values below:");
25
26         for (int i = 0; i < 10; i++) {
27             while (true) {
28                 try {
29                     System.out.print("Enter integer for index " + i + ": ");
30                     numbers[i] = input.nextInt();
31                     break;
32                 } catch (InputMismatchException e) {
33                     System.out.println("Wrong input! Please try entering this integer again.");
34                     input.next();
35                 }
36             }
37         }
38     }
39 }
```

```

40
41     }
42
43     while (true) {
44         try {
45             System.out.print("\nEnter an index to check its value: ");
46             int index = input.nextInt();
47
48             System.out.println("The value at index " + index + " is: " + numbers[index]);
49             break;
50         } catch (InputMismatchException e) {
51             System.out.println("Please enter a valid integer number for the index.");
52             input.next();
53         } catch (ArrayIndexOutOfBoundsException e) {
54             System.out.println("Error: Wrong index! Valid indices are only from 0 to 9.");
55         }
56     }
57
58     input.close();
59 }
60 }
```

Output ×



Arrays (run) ×

Arrays (run) #2 ×



run:

Enter 10 integer values below:

Enter integer for index 0: 1

Enter integer for index 1: 2

Enter integer for index 2: 3

Enter integer for index 3: 4

Enter integer for index 4: 5

Enter integer for index 5: 6

Enter integer for index 6: 7

Enter integer for index 7: 8

Enter integer for index 8: 9

Enter integer for index 9: 0

Enter an index to check its value: 4

The value at index 4 is: 5

BUILD SUCCESSFUL (total time: 15 seconds)

## Question no 2

```
Source History
package arrays;

/**
 *
 * @author PC
 */
import java.io.FileWriter;
import java.io.IOException;
import java.util.InputMismatchException;
import java.util.Scanner;

public class LaibaMudassarq2 {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);

        int choice = 0;
        double goldTola = 0;
        double silverTola = 0;
        double cashAmount = 0;
        double totalAssetsValue = 0;
        double zakatToBePaid = 0;
        String recordText = "";

        System.out.println("--- Zakat Calculator Menu ---");
        System.out.println("1. Gold Asset Only");
        System.out.println("2. Silver Asset Only");
        System.out.println("3. Cash Asset Only");
        System.out.println("4. Mixed Assets (Cash, Gold, and Silver)");

        while (true) {
            try {
                System.out.print("Select an option (1-4): ");
                choice = input.nextInt();
                if (choice >= 1 && choice <= 4) {
                    break;
                }
                System.out.println("Please enter a number between 1 and 4 only.");
            } catch (InputMismatchException e) {
                System.out.println("Invalid input type. Integers only.");
                input.next();
            }
        }

        switch (choice) {
            case 1:
                while (true) {
                    try {
                        System.out.print("How many Tola of Gold do you have? ");
                        goldTola = input.nextDouble();
                        break;
                    } catch (InputMismatchException e) {
                        System.out.println("Please enter a valid decimal or integer number.");
                        input.next();
                    }
                }

                if (goldTola >= 7.5) {
```

```

        if (goldTola >= 7.5) {
            totalAssetsValue = goldTola * 315000;
            zakatToBePaid = totalAssetsValue * 0.025;
        } else {
            System.out.println("Gold amount is below 7.5 Tola limit. No Zakat due.");
        }

        recordText = "Asset Type: Gold Only, Gold Tola: " + goldTola + ", Total Value: RS " + totalAssetsValue;
        break;

    case 2:
        while (true) {
            try {
                System.out.print("How many Tola of Silver do you have? ");
                silverTola = input.nextDouble();
                break;
            } catch (InputMismatchException e) {
                System.out.println("Please enter a valid decimal or integer number.");
                input.next();
            }
        }

        if (silverTola >= 52) {
            totalAssetsValue = silverTola * 3320;
            zakatToBePaid = totalAssetsValue * 0.025;
        } else {
            System.out.println("Silver amount is below 52 Tola limit. No Zakat due.");
        }
    }
}

```

```

    }
    while (true) {
        try {
            System.out.print("Enter your Gold amount (in Tola): ");
            goldTola = input.nextDouble();
            break;
        } catch (InputMismatchException e) {
            System.out.println("Please enter a valid number.");
            input.next();
        }
    }
    while (true) {
        try {
            System.out.print("Enter your Silver amount (in Tola): ");
            silverTola = input.nextDouble();
            break;
        } catch (InputMismatchException e) {
            System.out.println("Please enter a valid number.");
            input.next();
        }
    }

    double totalGoldCashValue = goldTola * 315000;
    double totalSilverCashValue = silverTola * 3320;
    totalAssetsValue = cashAmount + totalGoldCashValue + totalSilverCashValue;
    zakatToBePaid = totalAssetsValue * 0.025;

    recordText = "Asset Type: Mixed, Cash: RS " + cashAmount + ", Gold Tola: " + goldTola + ", Silver Tola: " + silverTola + ", Combin

```

```

        recordText = "Asset Type: Mixed, Cash: RS " + cashAmount + ", Gold Tola: " + goldTola + ", Silver
        break;
    }
}

```

```

System.out.println("\n--- Results Output ---");
System.out.println("Total Assessed Asset Value: RS " + totalAssetsValue);
System.out.println("Final Zakat amount to be paid: RS " + zakatToBePaid);

```

```

try {
    FileWriter fileWriter = new FileWriter("zakat_records.txt");
    fileWriter.write(recordText + " | Calculated Zakat to pay: RS " + zakatToBePaid);
    fileWriter.close();
    System.out.println("\nData successfully saved to file 'zakat_records.txt'!");
} catch (IOException e) {
    System.out.println("Error saving content to the file.");
}

```

```

input.close();
}

```

```

// While loop to
// case 1
// while (true) {
//     System.out.print("Enter your Gold amount (in Tola): ");
//     goldTola = input.nextDouble();
//     break;
// }
// case 2
// while (true) {
//     System.out.print("Enter your Silver amount (in Tola): ");
//     silverTola = input.nextDouble();
//     break;
// }
// Calculate total values and zakat
// double totalGoldCashValue = goldTola * 315000;
// double totalSilverCashValue = silverTola * 3320;
// totalAssetsValue = cashAmount + totalGoldCashValue + totalSilverCashValue;
// zakatToBePaid = totalAssetsValue * 0.025;
// Create record text
// recordText = "Asset Type: Mixed, Cash: RS " + cashAmount + ", Gold Tola: " + goldTola + ", Silver Tola: " + silverTola + ", Combined Value: RS " + totalAssetsValue;
// Write to file
// try {
//     FileWriter fileWriter = new FileWriter("zakat_records.txt");
//     fileWriter.write(recordText + " | Calculated Zakat to pay: RS " + zakatToBePaid);
//     fileWriter.close();
//     System.out.println("\nData successfully saved to file 'zakat_records.txt'!");
// } catch (IOException e) {
//     System.out.println("Error saving content to the file.");
// }
// Close input
// input.close();
}

```

Output - Arrays (run) #2 ×

```
run:
--- Zakat Calculator Menu ---
1. Gold Asset Only
2. Silver Asset Only
3. Cash Asset Only
4. Mixed Assets (Cash, Gold, and Silver)
Select an option (1-4): 2
How many Tola of Silver do you have? 2.3
Silver amount is below 52 Tola limit. No Zakat due.

--- Results Output ---
Total Assessed Asset Value: RS 0.0
Final Zakat amount to be paid: RS 0.0

Data successfully saved to file 'zakat_records.txt'!
BUILD SUCCESSFUL (total time: 14 seconds)
-
```